

TAHOUA, Niger

WMO: 610430

Lat: 14.876N

Lon: 5.265E

Elev: 386

StdP: 96.77

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.8	15.9	-13.3	1.2	26.2	-10.1	1.7	25.9	12.8	22.8	11.8	23.8	4.0	80	0.479

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	11.3	42.1	20.0	41.2	20.0	40.3	19.9	25.7	33.6	25.3	32.8	25.0	32.3	3.4	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.1	19.9	27.8	23.7	19.4	27.9	23.2	18.8	28.0	81.6	33.5	79.8	32.7	78.4	32.4	30.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.1	8.7	7.8	12.1	44.0	1.1	0.7	11.4	44.5	10.7	45.0	10.1	45.4	9.4	45.9		
(5)			4.4	28.4	0.6	1.5	4.0	29.5	3.7	30.3	3.3	31.2	2.9	32.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	29.6	23.7	26.7	30.6	34.1	34.6	32.6	30.2	28.8	30.2	31.3	28.4	24.4		
	DBStd	3.94	2.61	3.22	2.71	1.68	1.76	2.08	1.93	1.79	1.76	1.43	1.99	2.22		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD10.0	7164	426	467	637	722	762	678	625	582	607	659	551	448		
	CDD18.3	4122	167	234	379	472	503	428	367	324	357	401	301	189		
	CDH23.3	59163	1808	2995	5729	7972	8388	6557	4955	3952	4752	5926	4064	2065		
	CDH26.7	35859	790	1645	3624	5614	5922	4213	2694	1875	2575	3616	2351	940		
(14) Wind	WSAvg	3.9	4.7	4.4	4.3	3.5	3.8	4.1	4.0	3.6	2.9	2.9	3.6	4.4		
(15) Precipitation	PrecAvg	368	0	0	1	2	14	52	98	128	58	10	0	0		
	PrecMax	631	5	2	17	51	65	143	262	326	124	76	0	0		
	PrecMin	207	0	0	0	0	0	1	26	32	8	0	0	0		
	PrecStd	106	1	0	3	8	14	35	42	59	27	20	0	0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.0	39.9	41.0	43.1	43.2	41.7	38.7	37.0	39.1	39.8	38.2	35.7		
		MCWB	15.7	16.7	17.7	19.8	20.5	21.8	23.0	24.3	21.7	19.3	18.0	16.2		
	2%	DB	34.0	37.7	40.1	42.2	42.2	40.2	37.1	35.4	37.7	38.8	37.4	34.2		
		MCWB	14.9	16.4	17.4	19.4	20.4	22.7	23.6	24.3	22.6	19.6	17.7	15.6		
	5%	DB	32.5	36.2	39.2	41.3	41.3	39.1	35.9	34.2	36.4	38.0	36.7	33.0		
		MCWB	14.0	15.8	17.2	19.1	21.0	23.1	23.7	24.3	23.3	19.8	17.3	14.8		
	10%	DB	31.0	34.6	38.1	40.5	40.2	37.9	34.9	33.1	35.1	37.1	35.6	31.8		
		MCWB	13.5	15.1	16.7	18.9	21.3	23.2	23.6	24.2	23.7	19.8	16.7	14.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.7	19.6	21.9	25.0	26.6	26.2	25.7	25.9	26.1	25.4	23.5	17.6		
		MCDB	33.4	30.5	31.1	37.5	37.8	35.7	32.7	32.0	32.6	33.4	35.4	33.9		
	2%	WB	15.3	17.3	19.4	23.8	25.4	25.3	25.1	25.3	25.5	24.6	20.0	16.4		
		MCDB	32.9	35.8	37.0	36.4	36.1	34.5	32.0	31.5	32.1	32.4	33.1	32.9		
	5%	WB	14.4	16.2	18.1	22.9	24.6	24.8	24.7	25.0	25.1	24.0	18.0	15.4		
		MCDB	31.7	35.2	37.4	35.7	35.8	34.0	31.7	31.2	31.6	31.7	34.7	32.0		
	10%	WB	13.5	15.4	17.1	22.0	24.0	24.4	24.4	24.7	24.8	23.5	17.1	14.5		
		MCDB	30.4	34.1	37.0	35.1	35.3	33.7	31.5	30.7	31.3	31.5	34.6	30.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	14.0	13.6	12.8	11.3	10.1	9.1	8.5	9.9	12.1	14.1	14.1		
		MCDBR	15.1	15.0	14.5	13.8	12.7	11.4	10.3	9.8	11.5	13.1	14.5	15.2		
		MCWBR	6.8	6.5	6.0	5.7	5.1	3.4	2.7	2.5	3.5	4.7	5.6	6.6		
		MCDBR	14.9	14.5	13.8	11.1	10.9	10.3	9.2	8.9	9.9	11.0	13.9	14.5		
	5% WB	MCWBR	7.0	6.6	6.5	5.4	4.5	3.5	2.7	2.8	3.4	4.0	6.0	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.532	0.591	0.665	0.716	0.733	0.731	0.671	0.624	0.602	0.602	0.512	0.502		
	taud		1.802	1.637	1.504	1.474	1.472	1.513	1.669	1.753	1.776	1.704	1.808	1.841		
	Ebn at Noon		758	734	695	661	640	633	675	716	732	716	768	770		
	Edh at Noon		209	257	301	310	306	290	249	231	226	237	206	196		
(44) All-Sky Solar Radiation	RadAvg		5.49	6.03	6.50	6.69	6.58	6.30	6.24	6.07	6.35	6.12	5.72	5.33		
	RadStd		0.21	0.25	0.33	0.25	0.19	0.27	0.14	0.20	0.18	0.17	0.14	0.20		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page